Sanjaya Gebrial

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Vancouver, BC, Canada · Canadian citizen · TN-eligible for US roles

SUMMARY

Web developer with nearly 6 years in TypeScript and JavaScript on a small team building a production reading platform, strongest in rendering performance, pagination, and asynchronous code. Looking for a full-stack or front-end role.

SKILLS

Languages	TypeScript, JavaScript, HTML5, CSS3, Python, SQL, WGSL
Front-end	React, WebGPU, responsive layout, performance profiling
Back-end and data	Node.js, PostgreSQL, FastAPI
Tools	Git, GitHub Actions, CI/CD, Jest, Docker, BrowserStack

EXPERIENCE

Independent Software Engineering

Jan 2025 – Present

Self-directed engineering projects

Vancouver, BC · Began after a PressReader department restructure eliminated the role in December 2024

- Trained a 120M-parameter GPT-style language model from scratch and served it as a containerized FastAPI service with Postgres request logging and per-client rate limiting
- Built a browser path tracer on WebGPU by porting *Ray Tracing in One Weekend* from C to WGSL compute shaders, with a bounding volume hierarchy for faster ray traversal
- Fixed GPU random number generation whose output looked random but left visible structured artifacts in the Monte Carlo lighting
- Built an N-body simulation in TypeScript, using an octree to asymptotically speed up gravity calculations
- Wrote Jest tests for the octree's mass and acceleration results and for merging colliding bodies

stories.gebrial.ca · gebrial.ca/raytracing-webgpu · gebrial.ca/gravity

PressReader

Mar 2019 – Dec 2024

Software Engineer, Reading Platform

Richmond, BC · formerly Evident Point · TypeScript and Readium EPUB reader, 100,000+ titles

- Cut the time to open a book from roughly 10 seconds to 1.5 by profiling in Chrome DevTools, deferring off-screen spine items until the visible ones rendered, and prefetching only near the reader's position
- Designed the pagination model that keeps reading positions and bookmarks stable across devices and screen sizes, and built its Node pre-pass, which computes page breaks once in a headless browser
- Owned the read-aloud feature, moving it into the new reader, making its text processing asynchronous, and fixing the race that stopped speech when it outran processing
- Built React components for PressReader's web newspaper reader from design specs, and fixed rendering and cross-browser bugs that readers reported
- Fixed keyboard navigation in the ebook reader, including tabbing between links inside book content, and tested the reader on Android and iOS through BrowserStack
- Automated testing on every pull request with a GitHub Actions pipeline

Earlier: Software Developer Intern at Evident Point in 2017 and Grin Tech in 2016, Vancouver, BC.

EDUCATION

BSc, Combined Major in Computer Science and Physics, University of British Columbia 2018